

Vision Document

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The organization operates a large mobile app store that receives thousands of user reviews each week. Because of this high volume, manually sorting reviews into categories such as positive, negative, or neutral is not practical. To address this, the organization uses a machine learning system to analyze unstructured text (user reviews) and generate a predicted rating. Reviews are currently scored on a scale from 0 to 1 in increments of 0.1, where 0 represents the lowest rating and 1 represents the highest. To improve this process, the organization plans to implement a new software engineering lifecycle that better aligns with stakeholder needs and supports the development of a more reliable system for classifying reviews.

The current app review scoring system is functional but unreliable, producing scores that are frequently inaccurate and inconsistent. The underlying model is difficult to interpret, making it hard for users to understand how scores are generated or to have confidence in the results. The system also has no way to detect fake or manipulated reviews, leaving it open to exploitation and resulting in ratings that can be unfair or misleading. In many cases, predicted scores do not reflect the actual content of reviews, requiring developers and moderators to manually verify them which defeats the purpose of having an automated system in the first place. This undermines decision-making across the board. For example developers lack reliable signals for prioritizing fixes, and advertisers may invest in low-quality apps based on inaccurate scores. As a result, the system adds manual work, reduces operational efficiency, and cannot be counted on to deliver accurate insights.

The key stakeholders who will use the system are app store managers, app developers, and external partners such as app store advertisers. Each group will rely on the automated scoring system to inform their day-to-day decisions across a large volume of unstructured review data.

Following an evaluation of the current underperforming system, quality updates will be implemented to improve functionality and reliability in line with stakeholder requirements. The updated app scoring system will accept unstructured text as input and return a predicted

rating for each review, maintaining a minimum accuracy of 80% with a target range of 85–90%. The system will also incorporate filtering functionality to categorize reviews into three bins: positive, neutral, and negative, while sorting reviews on a more granular 1–10 scoring scale. Additional features will include a section highlighting the features with the most complaints and a section displaying the rating trend over time per app.

The updated system will introduce consistency and reliability, empowering stakeholders to make confident, data-driven decisions that ultimately benefit end users. For app store managers, the system provides clearer visibility into which apps are consistently underperforming and in need of further inspection, supporting decisions such as demotion or removal. For app developers, the complaint filtering and trend series features make it easier to pinpoint bugs and prioritize improvements within their apps. For external partners such as advertisers, the system helps identify the most reviewed and top-rated apps, making promotional decisions more targeted and informed. Overall, the updated system addresses the needs of all key stakeholders, delivering consistency, reliability, and efficiency across the board.

A reliable review scoring system is critical to maintaining the quality and trustworthiness of the app store, which directly impacts the organization’s reputation and the experience of its users. The updated system will be considered successful when predicted ratings consistently meet the 80% minimum accuracy threshold, scores are stable and consistent across similar reviews, and the reasoning behind ratings is interpretable enough for stakeholders to trust without manual verification. Additionally, the filtering and trend features should be actively used to drive app management, development, and advertising decisions. At that point, the system will have meaningfully reduced manual review work and restored confidence in the scoring process.